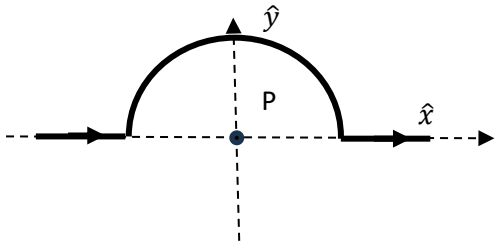
Physics 2 Problem Solving – Chapter 47 Boit Savart Law for Magnetic fields

1) Shown below is a semicircular thin wire with radius R and a uniform current flowing to the right. Draw and label on the diagram the incremental current, radius, angle, and any other variables needed to solve the problem.

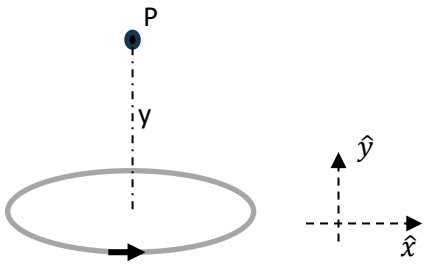
   Determine the magnitude and direction of the magnetic field at point P which is at the radius of the semicircle.

   $Answer$: $\vec{B} = \left(-\frac{\mu_o I}{4R}\right)\hat{z}$



2) A circular wire with a radius R and a constant current flowing counterclockwise viewed from the top is shown below. Determine the magnitude of the magnetic field at point P along the y-axis. Draw and label on the diagram the incremental current, radius, angle, and any other variables needed to solve the problem.

   $Answer$: $\vec{B} = \left(\frac{\mu_o I R^2}{2(R^2 + y^2)^{3/2}}\right)\hat{z}$



What happens to the equations for the magnetic field strength if the distance above the wire goes to zero (y = 0 m)?

$Answer$: $\vec{B} = \left(\frac{\mu_o I}{2R}\right)\hat{z}$

3) A rectangular wire with a constant CCW current I is shown below. Determine the magnitude of the magnetic field at point P above the end of the top wire. Assume s and L are constant and do not change in this problem. Draw and label on the diagram the incremental current, radius, angle, and any other variables needed to solve the problem. Ignore the constitutions to the magnetic field from the sides and bottom wire.

Answer:

$$Answer:\ \vec{B} = \left(\left[-\frac{\mu_o I}{4\pi s}\right]\left[\frac{L}{(L^2+s^2)^{1/2}}\right]\right)\hat{z}$$

